How-To Library

Informatica

Detecting Outliers in Column Profile Results in Informatica Analyst

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Abstract

When you analyze data in data sources, you might come across data that do not fall within an expected range of values or patterns, such as employee names that do not fall within an expected set of patterns or a Social Security Number that appears multiple times in an Employee database. To identify these anomalies, you can create a column profile on the data source and detect outliers in the profile results. This article describes how a data analyst can use outliers in a column profile results to identify anomalies in data patterns, values, and frequency in the data source.

Supported Versions

- Data Quality 10.0

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Overview

An outlier is a pattern, value, or frequency for a column that does not fall within an expected range of values. In Informatica Analyst, you can detect outliers in the summary view of profile results to identify patterns, values, and frequency that fall outside the expected range of values. You can view the outlier values in a column in the detailed view. Previously, you had to create multiple rules for the columns and use the rules in a column profile to identify anomalies in the profile results.

When you run the outliers, the profiling plug-in in the Data Integration Service runs the adjusted box plot algorithm to identify value-based, frequency-based, and pattern-based outliers in the profile results.

The adjusted box plot algorithm is similar to the interquartile range algorithm. The adjusted box plot algorithm uses the medcouple formula to compute the median difference between the left and right boundaries of data. After computing the left and right skewness of data or the decision boundaries, the adjusted box plot algorithm identifies the data points that fall beyond the boundaries as outliers.

When the adjusted box plot algorithm runs on the profile results, the algorithm identifies the data that falls below or above the decision boundary as outliers. The algorithm detects value-based, frequency-based, and pattern-based outliers in the profile results. The algorithm computes and identifies the values, frequency of values, and patterns in the data that fall below or above the decision boundary. The profiling plug-in uses the adjusted box plot algorithm because the algorithm is immune to extreme values and does not assume the normal distribution of data during the analysis of profile results.

You can view the following types of outliers in the summary view of profile results:

- Value frequency outliers. The algorithm computes the outliers based on the data type and its frequency in the column. The algorithm runs frequency-based outlier computation for columns with a non-numerical data type. For columns with a numerical data type, the algorithm runs value-based and frequency-based computation.
- Pattern outliers. The algorithm computes the outliers based on the data pattern and the frequency of the patterns in a column in the profile results. The algorithm runs frequency-based and pattern-based outlier computation to identify pattern outliers.

Business Example

You are a data analyst in an organization. Your organization has acquired a startup. Before you merge the data from both the organizations, you want to analyze the source data and metadata in the database of the startup company.

First, you need to analyze the data quality issues in the Employees table. To do this, create a profile and detect outliers in the Employees table.

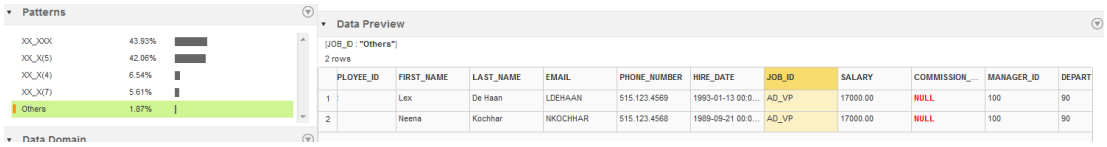
To view data quality issues and detect outliers, perform the following tasks:

1. In Informatica Analyst, create a physical data object from the Employees table.
2. Create a column profile on the data object.
3. Run the profile.
4. In the summary view, you can view the profile results for all the columns in the Employees table. Click a column to view the results for the column in detailed view.
5. To detect outliers, click **Actions** > **Detect Outliers**.
   The **Filter By** pane displays the number of columns that have pattern outliers and value frequency outliers.
6. Click the **Pattern Outlier** filter to view the columns with pattern outliers. Click the column to view the column results in detailed view.
7. Click he **Value Frequency Outlier** filter to view the columns with value frequency outliers. Click the column to view the column results in detailed view.

Pattern Outlier

In the Employees table, the pattern outlier algorithm identifies the column **JOB_ID** as a pattern outlier.

The following image shows a sample pattern outlier in the **Patterns** pane in detailed view:

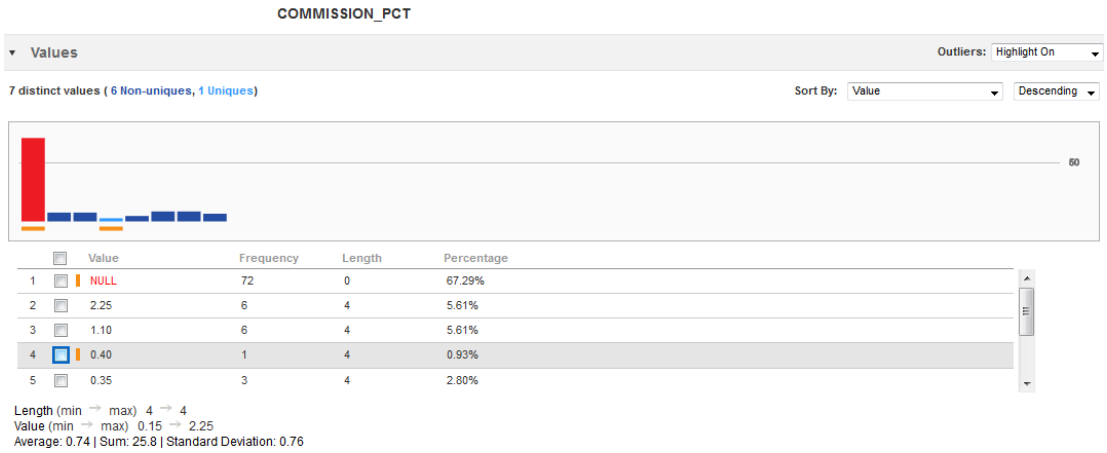


The pattern outlier identifies the JOB_ID **AD_VP** with 1.87% as the outlier as the value is outside the expected range of pattern and percentage. The algorithm uses patterns and their corresponding percentages to identify pattern outliers.

Value Frequency Outlier

In the Employees table, the value frequency outlier algorithm identifies the column **COMMISSION_PCT** as a value frequency outlier.

The following image shows a sample value frequency outlier in the **Values** pane in detailed view:



The value frequency outlier identifies the NULL value with frequency 72 and the 0.40 value with frequency 1 as the outlier. The algorithm uses values and their corresponding percentages to identify value frequency outliers.

Detecting Outliers

You can run outliers to identify patterns, values, or frequencies in a column that do not fall within an expected range of values.

1. In Informatica Analyst, click **Library** > **Assets** > **Profiles**.
2. Click **Actions** > **Open** to open the profile.

   The summary view of profile results appears in the **Discovery** workspace.
3. In the summary view, click **Actions** > **Detect Outlier**.

   The **Pattern outlier** and **Value Frequency outlier** in the **Filter By** pane changes from N/A to the number of columns that have outliers.
4. In the **Filter By** pane, click **Pattern outlier**.

   The columns with pattern outliers appear in the summary view.
5. In the **Filter By** pane, click **Value Frequency Outlier**.

   The columns with value frequency outliers appear in the summary view.
6. Click a column to view the detailed view of the column.
7. In the detailed view, select **Highlight On** from the outlier drop-down list.

   In the **Values** pane, the outliers appear as vertical bars with underlines in Orange color.
8. Click **Filter** in the **Outliers** list to view only the outlier values.

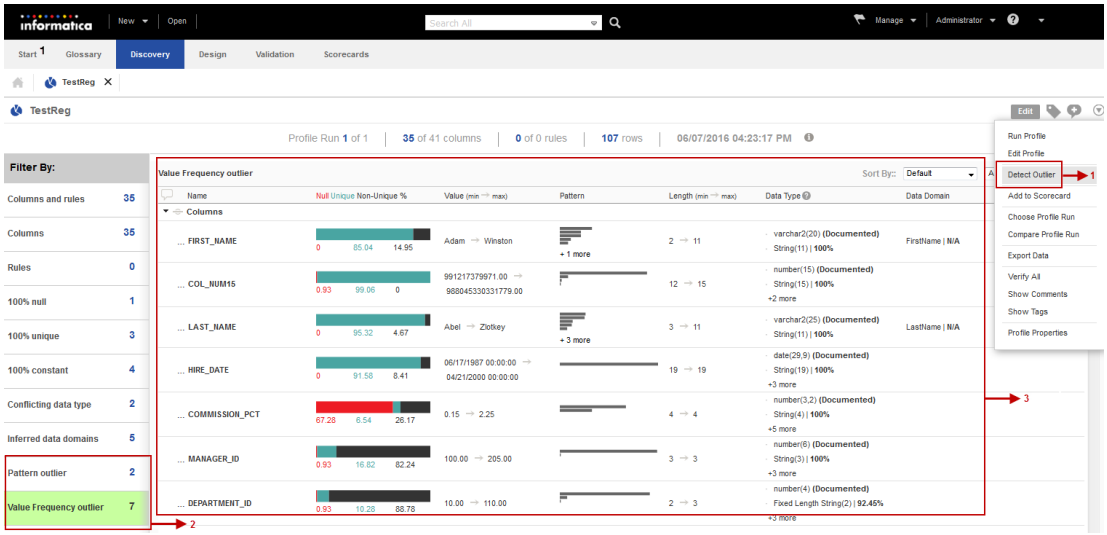
Summary View for Outliers

You can detect and view the outliers in the summary view of column profile results in Informatica Analyst. You can view the outliers after you run the outliers on the profile results.

You can click the following filters in the **Filter By** pane to view the outliers.

- Pattern outlier. The filter displays the outliers based on the patterns and pattern frequency in the column.
- Value Frequency outlier. The filter displays the outliers based on the values and value frequency in the column.

The following image shows a sample graphical view of summary view with outliers:



1. Detect Outlier option in Actions menu. You can click Detect Outlier in the Actions menu to detect outliers in the profile results.
2. Outlier filters in the Filter By pane. After you run outliers, the number of columns with outliers appear in the Filter By pane. When you click the filter, the columns pertaining to the filter appears in the summary view.
3. Summary view of profile results. You can view all the columns with outliers when you click the filter in the Filter By pane.

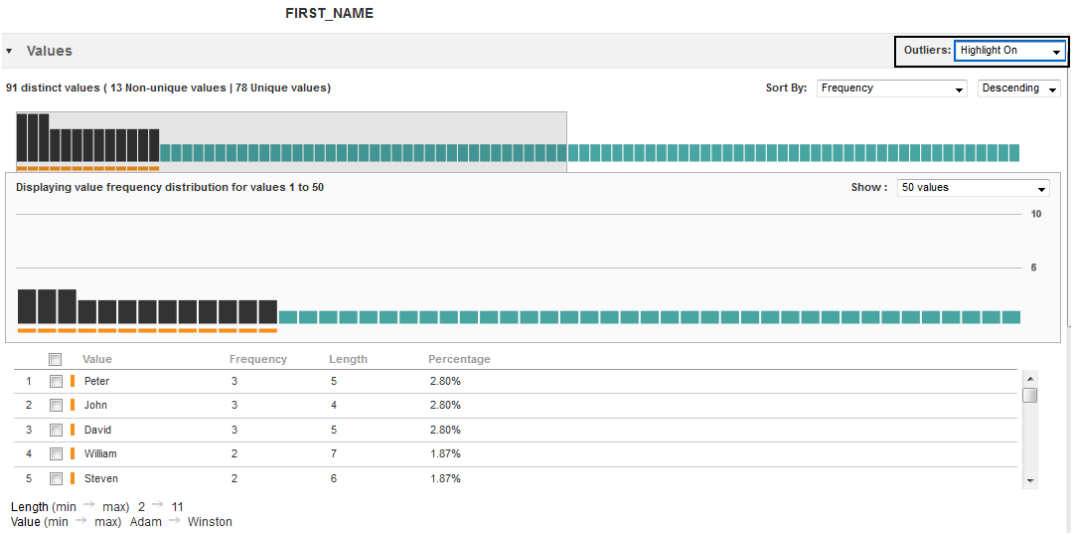
Detailed View for Outliers

You can view the outliers in a column in the detailed view.

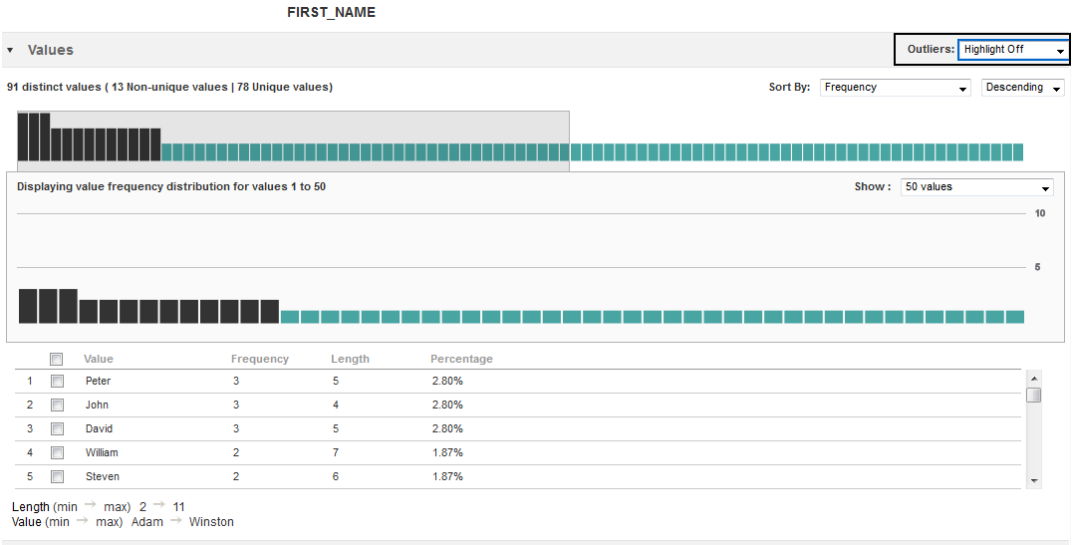
After you click a column in the summary view, the detailed view appears. You can view the outlier values in the **Values** pane as vertical bars with Orange underline. The Black vertical bars represent the frequency of a value in the column. You can use the **Outliers** list and **Sort by** list to view the outliers.

You can choose one of the following options in the **Outliers** list:

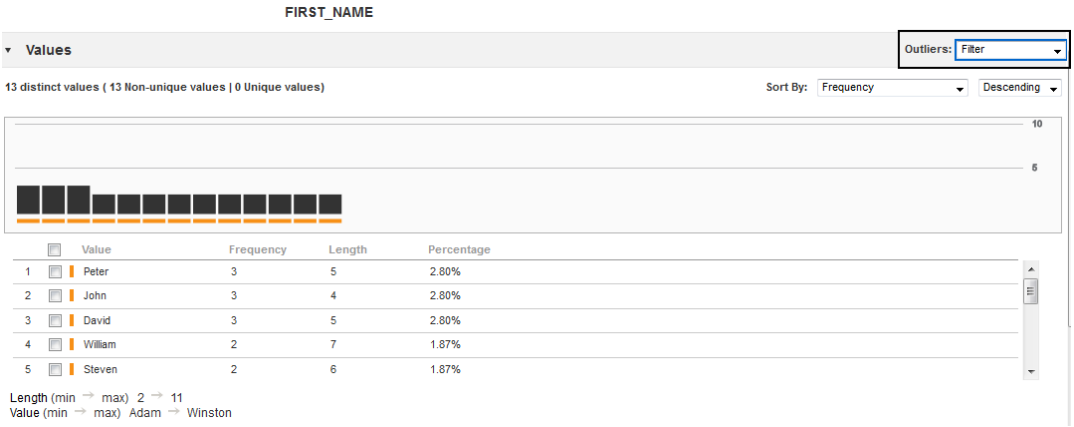
- Highlight On. When you choose the **Highlight On** option, you can view the outlier values in the column as vertical bars with Orange underline.
  The following image shows a sample graphical view of Values pane with Highlight On option:



- Highlight Off. When you choose the **Highlight Off** option, you can view the values in the columns and the outliers are not highlighted.
  The following image shows a sample graphical view of Values pane with Highlight Off option:



- Filter. When you choose the **Filter** option, you can view only the outliers in the column.
  The following image shows a sample graphical view of Values pane with Filters option:



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